

Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as “star-galaxy separation”, is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory’s Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide a binary classifier **extendedness** in each band, based on the threshold difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes. While this parameter provides a useful baseline for morphological classification, its binary nature prevents the optimal combination of information from multiple bands, as well as from other classification methods (e.g., those based on colors or external datasets). In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020), which overcomes these shortcomings. In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ and magnitude m_{CModel} . At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which, with appropriately chosen priors for the star-to-galaxy number ratio, allows us to compute p_S , the posterior probability that a source is unresolved. We apply this method to the Rubin DP1 and DP2 object catalogs and validate the resulting classifications using external HST labels in the COSMOS field. Using these labels as the ground truth, we quantitatively compare the performance of the p_S -based classifier with that of the Rubin **extendedness** parameter. We find that the p_S -based classifier consistently outperforms the Rubin **extendedness** baseline, with the improvement increasing as additional bands are incorporated. Approximately, the p_S classifier achieves the same classification performance at a magnitude limit about one magnitude fainter than the **extendedness** classifier. Depending on the scientific application, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

1. INTRODUCTION

The morphological classification of resolved and unresolved sources⁴, commonly referred to as “star-galaxy separation”, is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin’s Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies and the galaxies that do exist are obviously extended on the sky. Star-galaxy separation becomes increasingly harder at faint magnitudes, such as those probed by Rubin’s Legacy Survey of Space

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⁴ Note that some stars can be resolved, such as Asymptotic Giant Branch stars, while quasars and galaxies can be unresolved.

and Time (LSST; for details see [Ž. Ivezić et al. 2019a](#) and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller.

A variety of methods have been proposed in the literature for morphological classification of sources in images. [C. T. Slater et al. \(2020a\)](#) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star–galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline ([R. H. Lupton et al. 2002](#)), or SExtractor pipeline ([E. Bertin & S. Arnouts 1996](#)), are closely related both to each other, and to the Bayesian model odds ratio, first derived by [W. L. Seaborg \(1979\)](#). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best performance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier **extendedness** in each band, based on the threshold difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes, $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$. For unresolved sources, this Δm is vanishing within noise, with **extendedness**=0, while for resolved objects, selected by $\Delta m > 0.016$ mag, **extendedness**=1. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin’s Data Preview 1 catalogs, see Fig. 28 in [Vera C. Rubin Observatory Team et al. \(2026\)](#).

Although **extendedness** was successfully used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star–galaxy separation threshold is optimized at the faint, low S/N, end of the data, the ability to recognize barely resolved objects at the bright end is not fully exploited. More importantly, the binary nature of **extendedness** parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands. With appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved), can be computed in Bayesian framework.

The main goal of this paper is to implement the steps towards Bayesian star–galaxy separation outlined in SIL2020 using Rubin’s Data Preview 2 data. We describe our dataset and method in §2, and quantify its performance using HST-based labels in §3. We discuss our results and how to retrieve our catalog with improved star–galaxy separation in §4.

2. DATA AND METHODOLOGY

We first describe Rubin and HST data used in our analysis, followed by a brief review of the Bayesian star–galaxy separation method proposed in SIL2020. We conclude this Section with a description of detailed method implementation steps, and discuss the performance of Rubin’s default method and our Bayesian star–galaxy separation method in the next Section.

2.1. Rubin Data Preview 2

All data collected with the LSST Camera (XXX: REF) during the commissioning of the Rubin Observatory will be released as Data Preview 2. Coadded deep images and resulting catalogs have been released in July 2026. These coadded images are based on about 24,000 single-visit images and cover about 3,000 sq. deg. of sky. In our analysis we focus on the COSMOS field because it is both already well observed by Rubin (to about 1 magnitude shallower limit than anticipated for 10-year LSST) and space-based HST COSMOS2020/FARMER star and galaxy labels are available ([J. R. Weaver et al. 2022a](#)). As a control sample for code robustness, we also use Rubin data for the Extended Chandra Deep Field South (ECDFS).

For the COSMOS validation sample, we select Rubin DP2 sources in the rectangular region $149.7 < \text{RA} < 150.7$ and $1.7 < \text{Dec} < 2.7$, with $16 \leq r_{\text{CModel}} < 26$. Objects listed in the COSMOS2020/FARMER reference catalog are matched to DP2 sources using a 0.5 arcsec matching radius. Within the selected region and magnitude ranges, the DP2 sample contains 269,606 objects, of which 193,586 are matched to the COSMOS2020/FARMER objects and 76,020 are unmatched (a matching fraction of 72%). This somewhat low matching fraction is due to large areas around bright stars that are masked in the COSMOS2020/FARMER

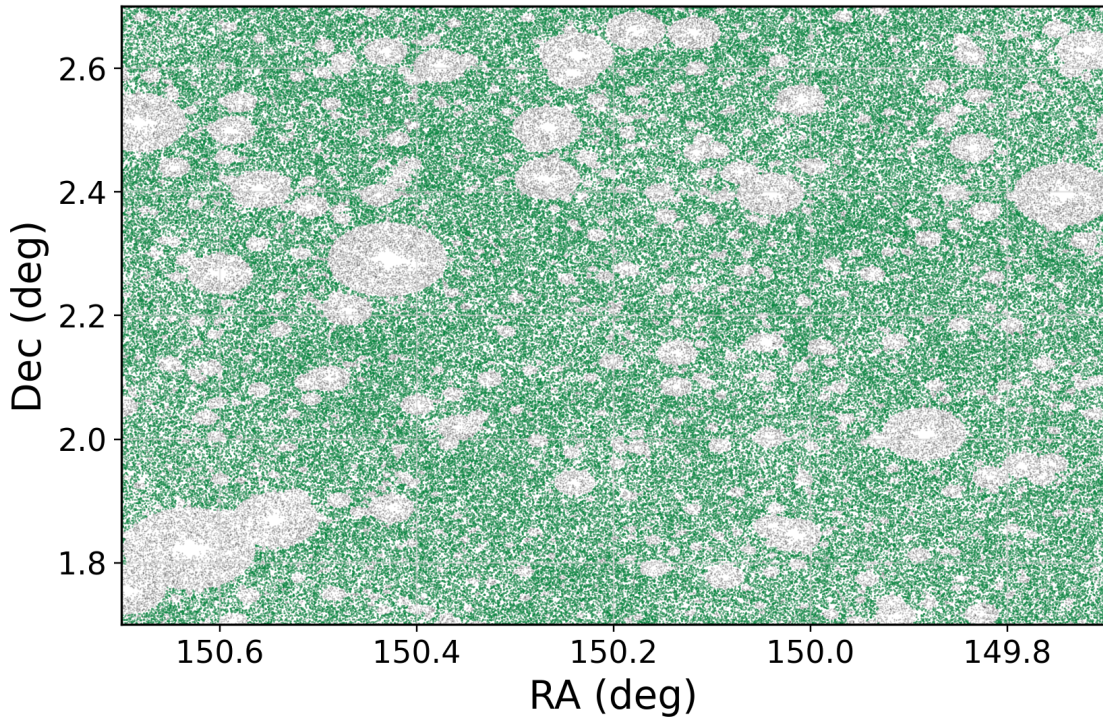


Figure 1. The footprint for the DP2-COSMOS sample analyzed here. Gray points show Rubin DP2 objects and green points show the subset matched to the COSMOS2020/FARMER sample from HST using a 0.5 arcsec matching radius.

catalog (see Figure 1). The matched validation sample contains 8,788 stars and 184,798 galaxies, classified using `truth_binary = 1` for stars and `truth_binary = 0` for galaxies. Their number counts as function of magnitude are shown in Figure 2. The low fraction stars is due to very faint sample limit and high Galactic latitude.

2.2. The distributions of sources classified using HST labels

The external COSMOS2020/FARMER labels provide the reference classification, considered here as ground truth, for examining the distribution of stars and galaxies in various diagrams and for measuring the performance of star-galaxy separation methods. Nevertheless, it is quite possible that some very faint sources classified as stars might be unresolved galaxies even at HST’s resolution.

The top left panel in Figure 3 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ vs. m_{CModel} diagram. As we describe in more detail below, modeling this distribution is a key ingredient of the Bayesian star-galaxy separation method developed here.

The top right panel in Figure 3 shows that galaxies, in addition to outnumbering stars at faint magnitudes, have on average SLIGHTLY bluer colors than stars. Color-color diagrams shown in the bottom two panels in Figure 3 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 4 and the top left panel in Figure 3 clearly show that increased noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods. It is notable that the top right panel in Figure 4 implies that most unresolved faint blue sources have $u - g$ color more typical of low-redshift quasars than of main sequence stars.

2.3. The distributions of sources classified using *extendedness* parameter

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 5, which compares subsamples classified using HST ground-truth labels and the default Rubin *extendedness* binary class in the r band. As evident, a significant fraction of faint sources within 2 magnitudes from the faint limit are misclassified when using default *extendedness* parameter. Misclassifications have a more severe impact

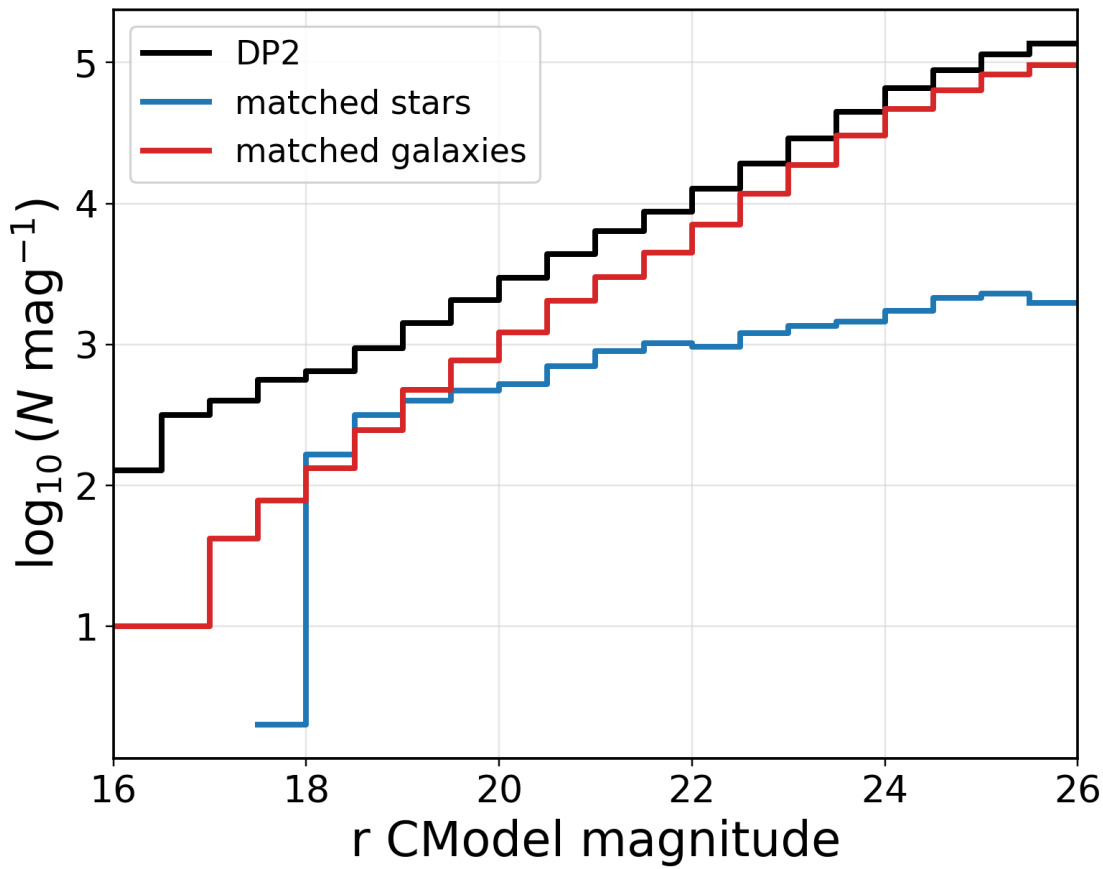


Figure 2. Magnitude distribution of the DP2-COSMOS sample. The black curve shows all selected DP2 sources, while the blue and red curves show matched COSMOS2020/FARMER sources classified as stars and galaxies, respectively. Counts are shown per magnitude, using 0.5 mag wide bins.

on stellar sample: only 26% of “stars” selected by `extendedness=1` condition are unresolved in HST data. This high level of contamination increases towards the faint end (see the bottom left panel in Figure 5). As the same time, the selection completeness for objects unresolved by HST is only 67%. Due to much higher counts, the purity and completeness for galaxies are much higher: 98% and 89%, respectively.

2.4. Bayesian Star–Galaxy Separation

Following SIL2020, we separate the evidence supplied by an individual morphology measurement from the expected relative abundance of the two source populations. Let U and R denote the unresolved and resolved hypotheses, and let d_b denote the morphology measurement in band b . At magnitude m , Bayes’ theorem gives the posterior odds

$$\frac{P(U \mid d_b, m)}{P(R \mid d_b, m)} = \frac{\pi_U(m) \mathcal{L}_{U,b}(d_b \mid m)}{\pi_R(m) \mathcal{L}_{R,b}(d_b \mid m)}.$$

Here $\mathcal{L}_{U,b}$ and $\mathcal{L}_{R,b}$ are the class-conditional morphology densities. Their ratio measures how strongly the observed morphology favors the unresolved or resolved hypothesis. The factors π_U and π_R are the expected fractions of the two populations at that magnitude before the morphology of the individual source is used. We denote the resulting unresolved-source posterior by

$$p_S \equiv P(U \mid d_b, m).$$

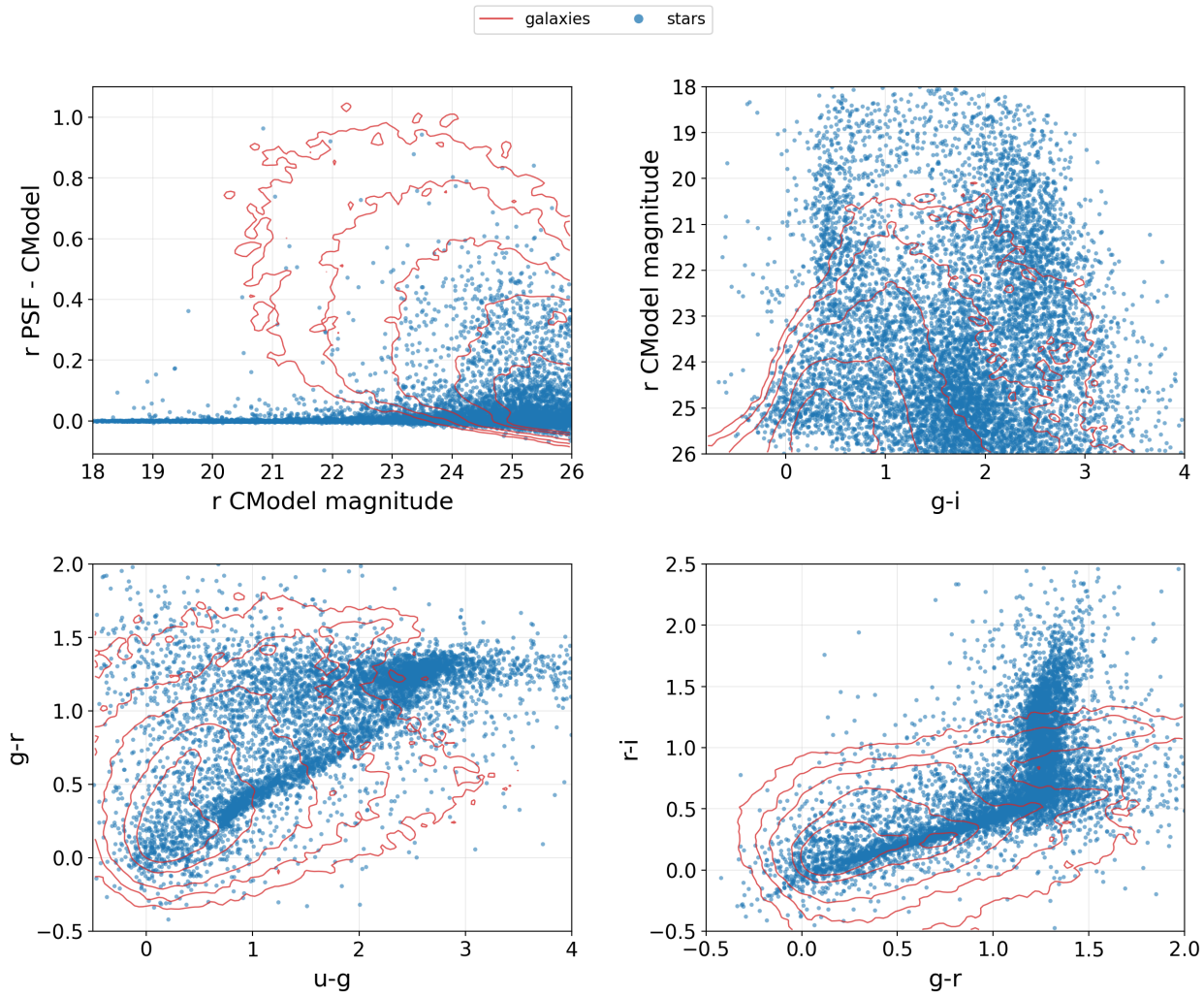


Figure 3. The distribution of stars (blue dots) and galaxies (red contours), classified using HST labels, in various informative diagrams. The r -band CModel magnitude is uncorrected for the ISM dust extinction, while the CModel-based colors are not. The top left panel shows the $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ vs. m_{CModel} diagram, where stars form a narrow sequence centered on $\Delta m \approx 0$ and for galaxies $\Delta m > 0$. The top right panel shows the r vs. $g - i$ color magnitude diagram and two bottom panels show representative color-color diagrams. In particular, unresolved sources with $u - g < 0.6$ are expected to be a mixture of low-redshift quasars, white dwarfs and unresolved binaries. XXX remove the title (here and elsewhere - the journal editors don't like it).

The subscript S indicates the star-like, unresolved component; it does not assert that every unresolved source is physically a star. If $O_{\text{U/R}}$ denotes the posterior odds in the preceding equation, then

$$p_{\text{S}} = \frac{O_{\text{U/R}}}{1 + O_{\text{U/R}}}.$$

For conditionally independent measurements in a set of bands $\mathcal{B}$, the morphology likelihood ratios may be combined as

$$\mathcal{B}_{\text{morph}} = \prod_{b \in \mathcal{B}} \frac{\mathcal{L}_{\text{U},b}(d_b | m_b)}{\mathcal{L}_{\text{R},b}(d_b | m_b)}.$$

In the multiband implementation, the r -band source-count prior is applied once after the usable per-band morphology evidence has been combined:

$$O_{\text{U/R}} = \frac{\pi_{\text{U}}(m_r)}{\pi_{\text{R}}(m_r)} \mathcal{B}_{\text{morph}}.$$

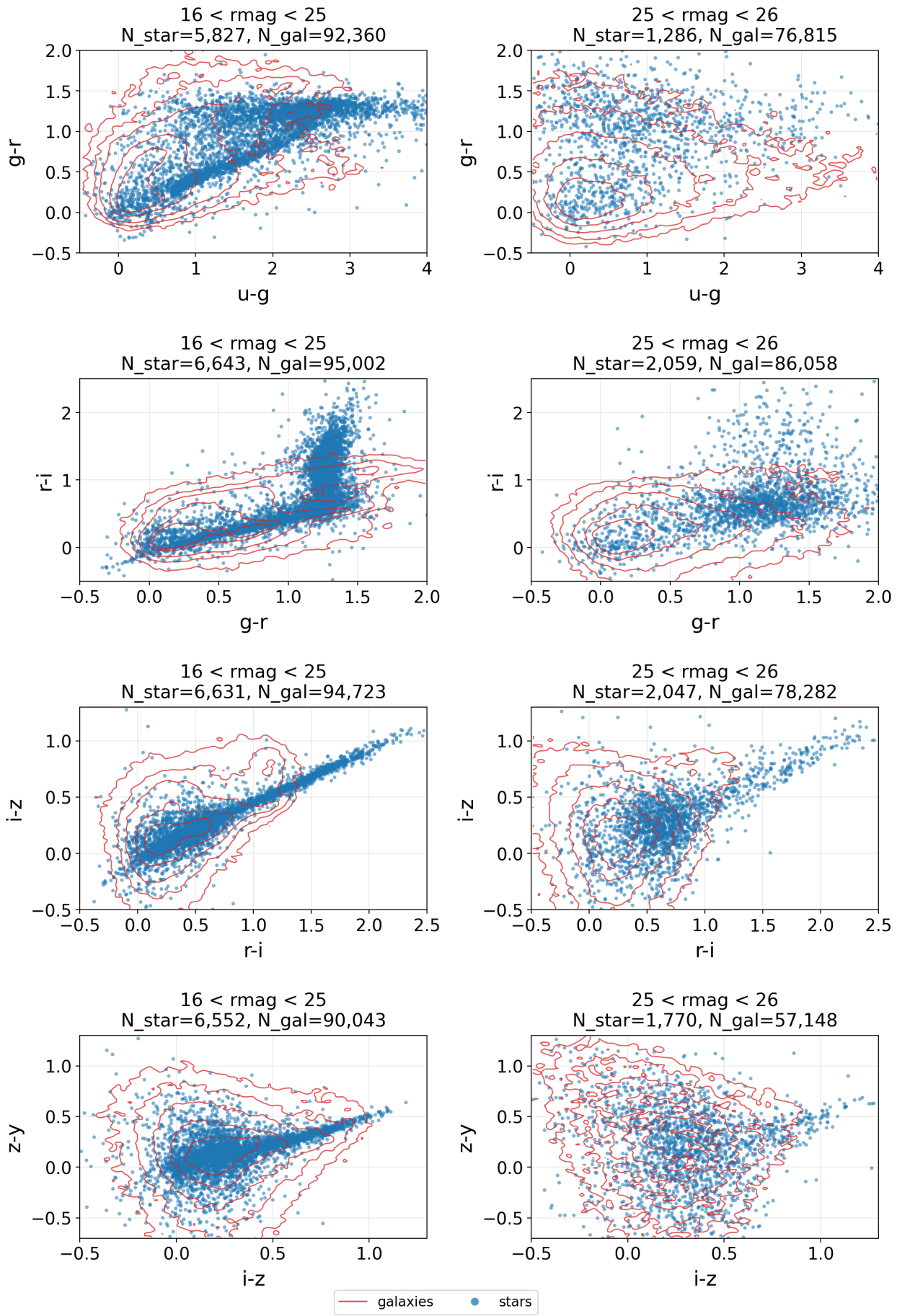


Figure 4. Similar to the bottom two panels in Figure 3, except that here all color-color diagrams are shown and data are split by r-band CModel magnitude to show deterioration close to the faint magnitude limit (left: $16 < r_{\text{mag}} < 25$; right: $25 < r_{\text{mag}} < 26$).

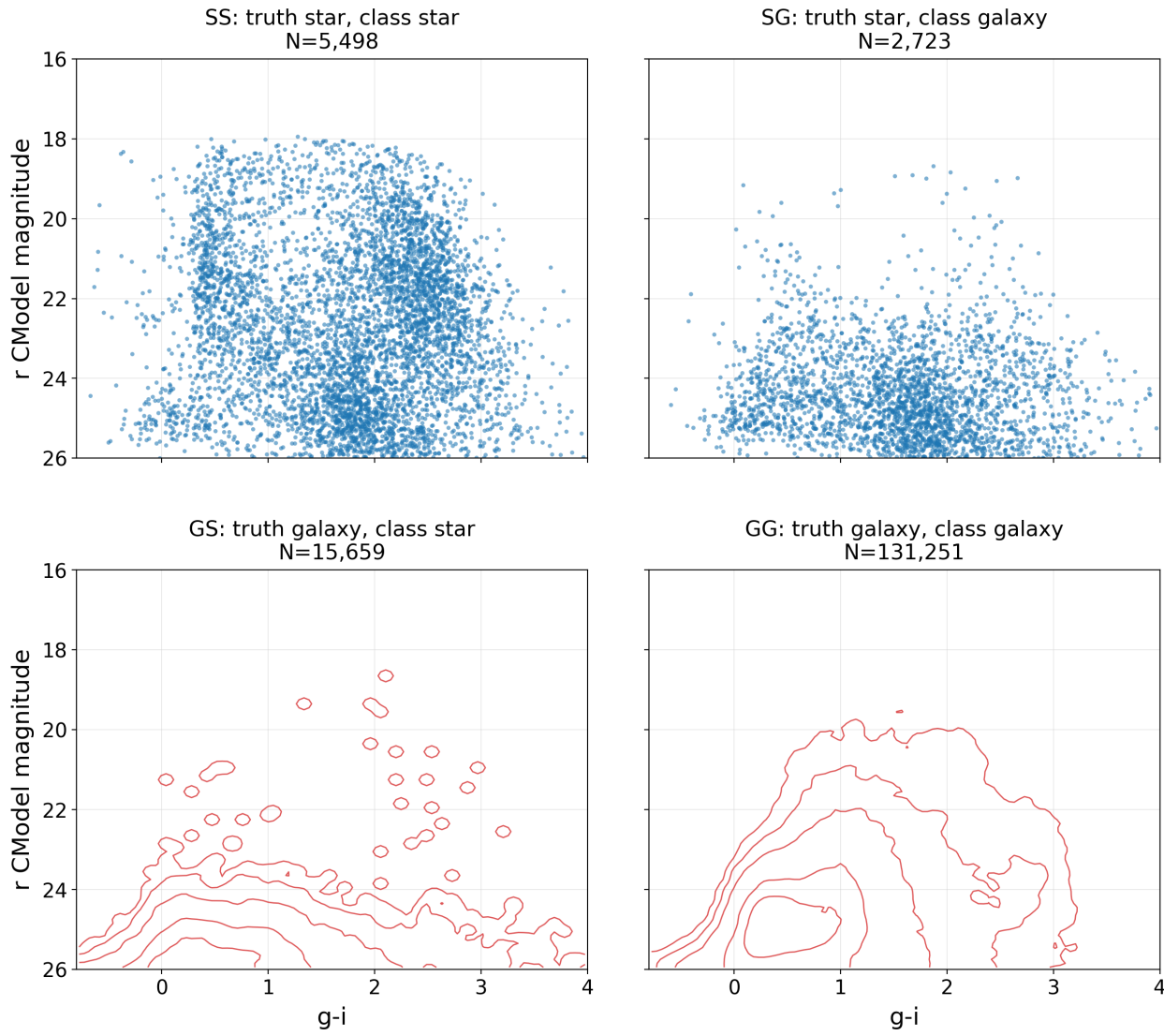


Figure 5. Performance of Rubin’s default **extendedness** parameter in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. $g - i$ color-magnitude diagram, with counts shown in the title above each panel. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by **extendedness**=0 is severely contaminated by galaxies for $r_{\text{CModel}} > 24$.

Applying the population prior once prevents the same source-count information from being counted repeatedly across bands. The product above treats the per-band morphology measurements as conditionally independent given the source class. A complementary classifier can be included in the same framework when its calibrated output can be expressed as a compatible likelihood ratio (M. Henrion et al. 2011; C. T. Slater et al. 2020b).

The continuous p_S value is the primary output of the method. When a binary catalog assignment is required, we use the maximum-posterior rule: objects with $p_S \geq 0.5$ are assigned to the unresolved class and objects with $p_S < 0.5$ to the resolved class. This convention is fixed and used for every binary classification reported in this paper.

2.5. Implementation of the Probabilistic Morphology Model

2.5.1. Mixture Model for Δm vs. m_{CModel} diagram

At a fixed CModel magnitude, the distribution of Δm is modeled as a mixture of an unresolved (star) component and K resolved (galaxy) components. The unresolved component is a single Gaussian with mean μ_U , width σ_U , and weight w_U . Each resolved component is a skew-normal distribution with weight $w_{R,k}$, location $\mu_{R,k}$, scale $\sigma_{R,k}$, and shape parameter $\alpha_{R,k}$. The full mixture probability density function is

$$p(\Delta m | m) = w_U \mathcal{G}(\Delta m; \mu_U, \sigma_U) + \sum_{k=1}^K w_{R,k} \mathcal{S}(\Delta m; \mu_{R,k}, \sigma_{R,k}, \alpha_{R,k}), \quad (1)$$

where $\mathcal{G}$ denotes the Gaussian and $\mathcal{S}$ the skew-normal distribution, and the weights are constrained to sum to unity: $w_U = 1 - \sum_k w_{R,k}$.

The fitting is performed independently in bins of CModel magnitude. At bright magnitudes ($m_{\text{CModel}} < 22$), where the resolved population is relatively simple, we use $K = 2$ resolved components in bins of width 1–3 mag spanning $15 < m_{\text{CModel}} < 22$. At faint magnitudes ($m_{\text{CModel}} \geq 22$), where the galaxy morphology distribution is more complex, we use $K = 3$ resolved components in finer bins of width 0.5–1 mag spanning $22 \leq m_{\text{CModel}} < 27$. Within each magnitude bin, Δm values in the range $(-0.5, 1.5)$ are histogrammed into 300 bins and normalized to a probability density. The mixture model is fit to these histograms using bounded least-squares optimization with Poisson-like uncertainties $\sigma_i = \sqrt{N_i + 1}/(N \delta)$, where N_i is the count in bin i , N is the total number of sources in the magnitude slice, and δ is the histogram bin width.

This fitting procedure is carried out in three steps. In the first step, all mixture parameters are fit independently in each magnitude bin. This free fit works well at bright magnitudes, where unresolved and resolved sources are clearly separated in Δm . However, at faint magnitudes ($m_{\text{CModel}} \gtrsim 24$), photometric noise broadens the unresolved Gaussian until it becomes degenerate with the resolved components. The fitter can no longer reliably distinguish the two populations, causing the unresolved Gaussian to absorb flux from the galaxy distribution. This inflates the inferred star fraction at faint magnitudes.

To prevent this degeneracy, we impose smooth constraints on the unresolved component. In the second step, the per-bin star parameters from step 1 are replaced with smooth functions of magnitude: μ_U is fixed to a single value (the median across all bins, typically ≈ 0 in each band); $\sigma_U(m)$ is fit with the photometric error model from B. B. Sesar et al. (2007) (their eq. 5),

$$\sigma_U(m) = \sqrt{\sigma_0^2 + a^2 \cdot 10^{b(m-25)}}, \quad (2)$$

where σ_0 , a , and b are free parameters; and $w_U(m)$ is modeled as a linear fit to $\log_{10}(w_U)$ versus magnitude, using only bins brighter than $m_{\text{CModel}} = 24$ where the free fit is reliable. This smooth model is then extrapolated to fainter magnitude bins. In the third step, with the unresolved component (μ_U, σ_U, w_U) fixed to the smooth curves from step 2, only the resolved skew-normal parameters are re-fit in each magnitude bin. The resolved weights are rescaled to sum to $(1 - w_U)$ so that the total mixture remains a proper probability density. The effectiveness of this constrained fitting procedure is demonstrated in Figure 6, which compares the galaxy-to-star count ratio $\log_{10}(N_{\text{gal}}/N_{\text{star}})$ as a function of magnitude for models with and without the smooth constraints. The unconstrained model exhibits a spurious turnover at faint magnitudes, while the constrained model recovers the expected monotonically increasing ratio in agreement with the COSMOS2020 external truth labels.

From the fitted model, the probability that a source is unresolved is computed as the ratio of the unresolved component to the total mixture density at the observed $(\Delta m, m_{\text{CModel}})$:

$$p_S = \frac{w_U \mathcal{G}(\Delta m; \mu_U, \sigma_U)}{p(\Delta m | m)}. \quad (3)$$

The model parameters are looked up by the CModel magnitude bin in which each source falls. The quality of the fits is illustrated in Figure 7, which shows the mixture model overlaid on the observed Δm histograms in each magnitude bin, and in Figure 8, which shows the 2D data, model, and residual maps in the Δm vs. m_{CModel} plane.

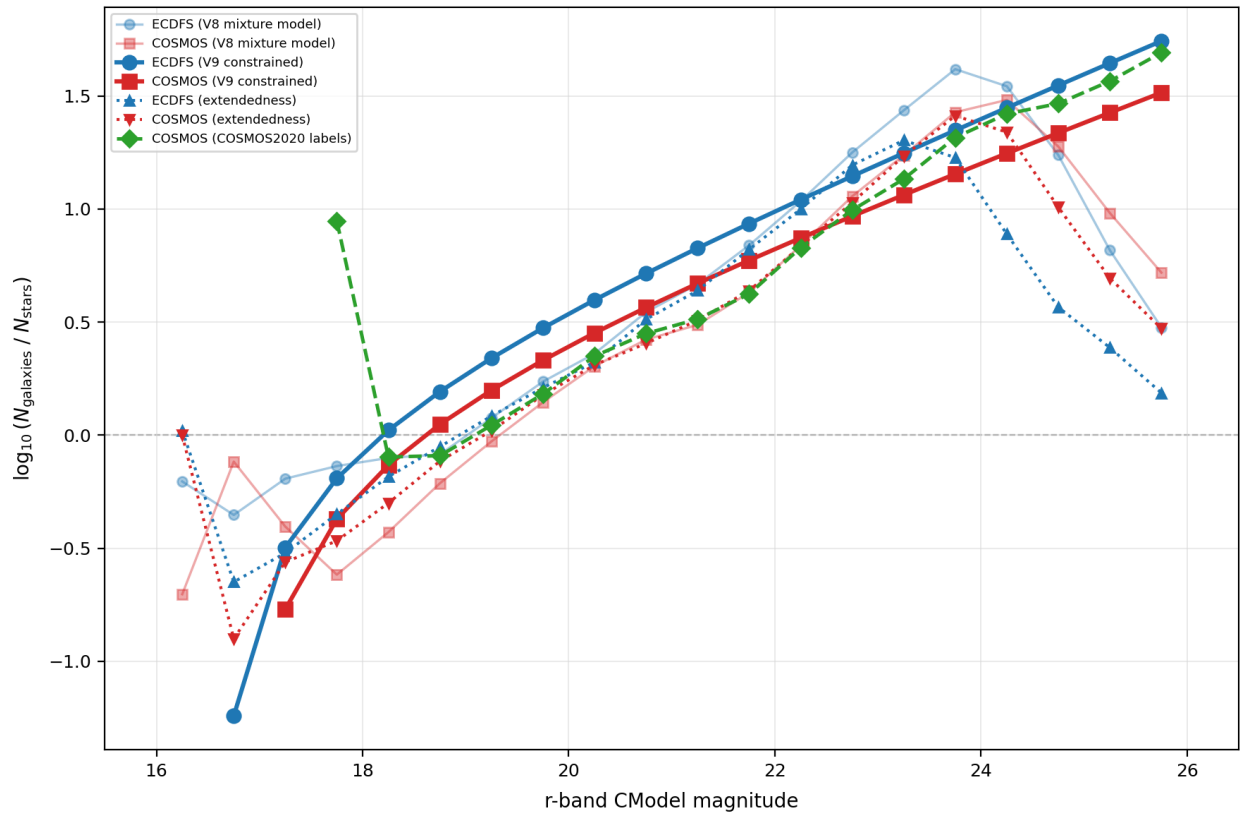


Figure 6. Galaxy-to-star count ratio $\log_{10}(N_{\text{gal}}/N_{\text{star}})$ vs. r-band CModel magnitude. The unconstrained mixture model (V8, faded curves) exhibits a spurious turnover at $m \gtrsim 24$ due to the unresolved Gaussian absorbing galaxy flux. The constrained model (V9, bold curves) recovers the expected monotonically increasing ratio, in good agreement with the COSMOS2020 truth labels (green dashed). Dotted lines show the Rubin **extendedness**-based classification for comparison.

2.5.2. Magnitude-dependent Source-count Priors

The normalization parameters of the best-fit mixture provide a magnitude-dependent prior for the relative abundance of unresolved and resolved sources. Here “prior” means the expected class ratio at a given magnitude before the morphology measurement of an individual source is considered. It is therefore a population term, rather than an alternative choice of the $p_S = 0.5$ catalog-assignment boundary. In the notation of Section 2.4, the fitted priors are

$$\pi_U(m) = w_U(m), \quad \pi_R(m) = \sum_{k=1}^K w_{R,k}(m) = 1 - w_U(m). \quad (4)$$

The corresponding resolved-to-unresolved count ratio is

$$R_{R/U}(m) \equiv \frac{N_R(m)}{N_U(m)} = \frac{\pi_R(m)}{\pi_U(m)} = \frac{1 - w_U(m)}{w_U(m)}. \quad (5)$$

When the unresolved and resolved populations are identified with stars and galaxies, respectively, $R_{R/U}$ is the galaxy-to-star count ratio shown in Figure 6. Equation 3 already incorporates this prior through w_U and the total mixture density. The prior must therefore be applied only once; multiplying an additional count ratio into Equation 3 would double-count the same population information.

For comparison, the HST-based external COSMOS ratio is measured in the same magnitude bins as

$$\hat{R}_{G/S}^{\text{HST}}(m_j) = \frac{N_{\text{gal,HST}}(m_j)}{N_{\star,\text{HST}}(m_j)}, \quad (6)$$

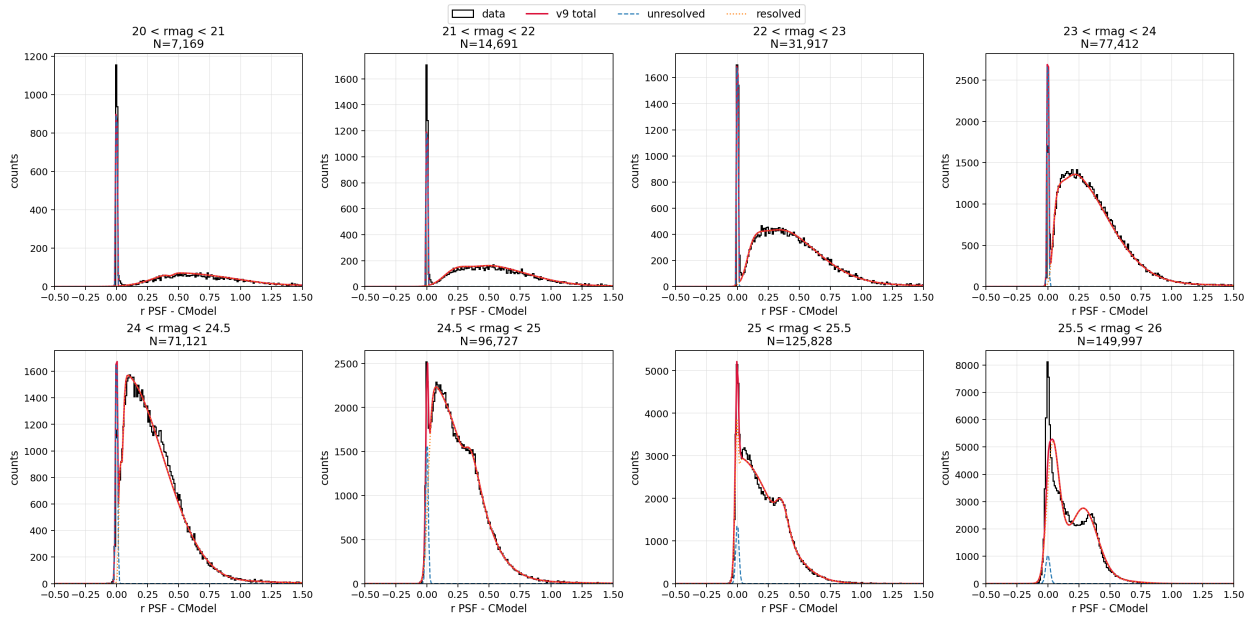


Figure 7. Observed Δm histograms (black) and fitted mixture models (red) for the r-band in each CModel magnitude bin. The unresolved Gaussian component is shown as a blue dashed line and the resolved skew-normal components as orange dotted lines.

using the matched COSMOS classifications described in Section 2.1 (A. Leauthaud et al. 2007; A. C. Robin et al. 2007; J. R. Weaver et al. 2022b). Because these labels do not enter the mixture fit, the HST curve is an external empirical check on the inferred abundance trend. The agreement of the constrained curve with the overall rise of the HST ratio supports the use of the fitted normalization as a prior. The statistical uncertainty of this ratio increases in bins containing few reference stars.

TRILEGAL provides a complementary prediction for the stellar surface density along the COSMOS line of sight (L. Girardi et al. 2005; P. Dal Tio et al. 2022). Because TRILEGAL predicts stellar counts rather than galaxy counts, we compare it directly with the stellar surface density in the COSMOS reference sample. In magnitude bin m_j , the observed stellar surface density is

$$\hat{n}_{\star}^{\text{COSMOS}}(m_j) = \frac{N_{\star, \text{COSMOS}}(m_j)}{A_{\text{eff}} \Delta m_j}, \quad (7)$$

where A_{eff} is the effective matched area and Δm_j is the bin width. The COSMOS and TRILEGAL counts are compared using the same photometric band, extinction convention, magnitude bins, source selection, and solid-angle normalization. This direct comparison avoids introducing an empirical galaxy numerator into the TRILEGAL diagnostic. The posterior probabilities used in this work retain the empirical mixture prior of Equation 4; neither the HST count ratio nor the TRILEGAL stellar-count prediction is used to modify the posterior or the fixed $p_S = 0.5$ catalog-assignment boundary.

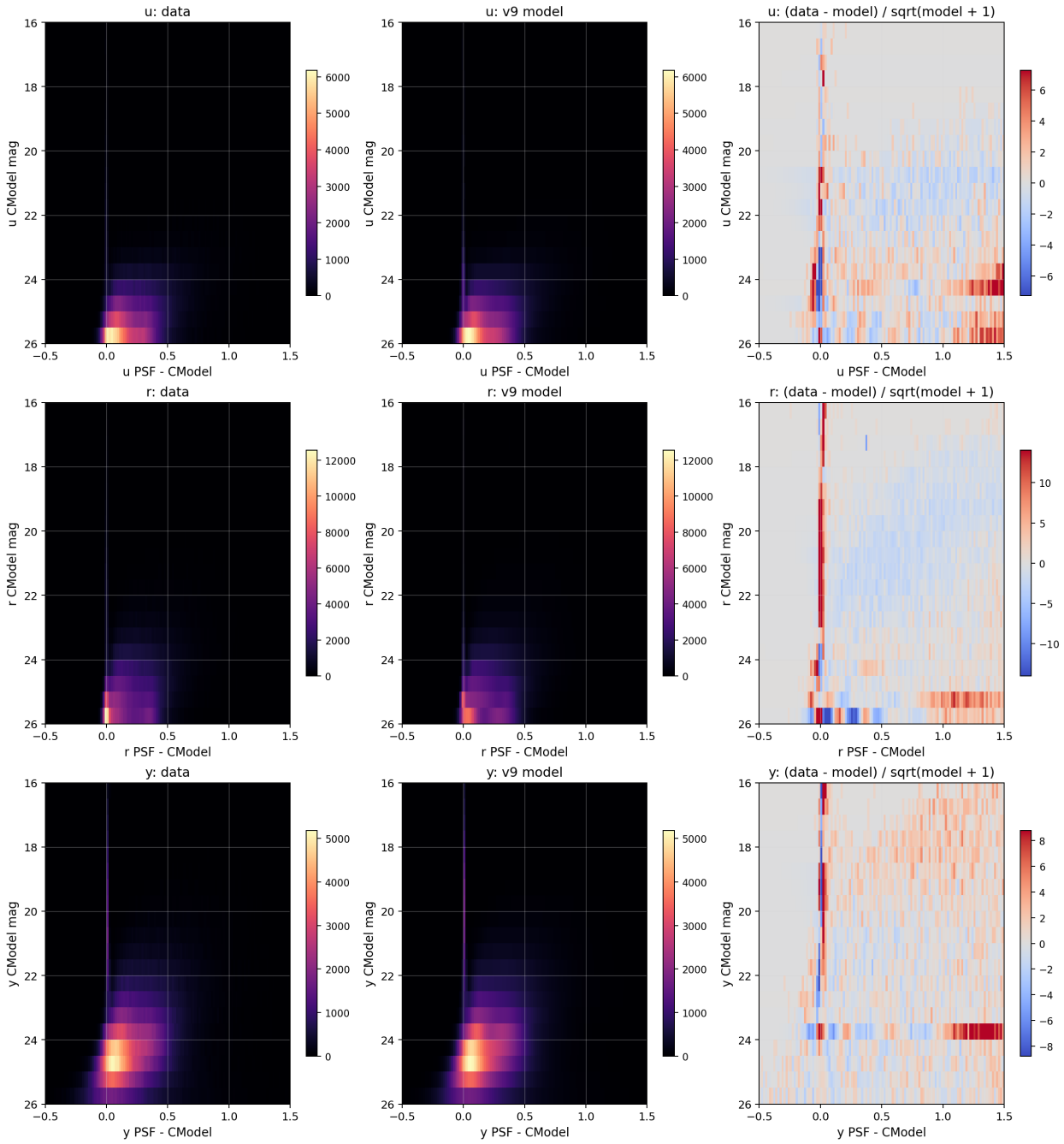


Figure 8. Two-dimensional comparison of the observed (left), modeled (center), and residual (right) source density in the Δm vs. m_{CModel} plane. The residual panels show the localized and magnitude-dependent differences that remain between the constrained mixture model and the data.

3. PERFORMANCE OF THE BAYESIAN CLASSIFIER

3.1. *Validation Metrics*

We evaluate the classifiers using the matched COSMOS2020/FARMER reference sample (J. R. Weaver et al. 2022b). The one-to-one match, constructed with a maximum separation of $0.5''$, contains 193,586 objects: 8,788 reference-labeled stars and 184,798 reference-labeled galaxies. Throughout this section, r denotes the Rubin r -band CModel magnitude.

The validation sample becomes strongly galaxy dominated at faint magnitudes. An overall accuracy would consequently be uninformative: a classifier could obtain a high value by assigning nearly every faint source to the galaxy class while recovering few stars. We therefore evaluate the star-candidate and galaxy-candidate samples separately. For a target class X , where X is either stars or galaxies, we define completeness, purity, and contamination as

$$C_X \equiv \frac{\text{TP}_X}{\text{TP}_X + \text{FN}_X}, \quad (8)$$

$$P_X \equiv \frac{\text{TP}_X}{\text{TP}_X + \text{FP}_X}, \quad (9)$$

$$f_{\text{cont},X} \equiv 1 - P_X = \frac{\text{FP}_X}{\text{TP}_X + \text{FP}_X}. \quad (10)$$

Completeness is the fraction of reference objects of class X that are recovered, whereas purity is the fraction of the selected sample that belongs to class X according to the reference labels. Purity is particularly important for the imbalanced faint sample because even a modest false-positive rate can produce substantial contamination (T. Saito & M. Rehmsmeier 2015).

For a receiver operating characteristic (ROC) curve, the false-positive rate for target class X is

$$\text{FPR}_X \equiv \frac{\text{FP}_X}{\text{FP}_X + \text{TN}_X}. \quad (11)$$

Its denominator contains all reference members of the opposite class and therefore differs from the denominator of contamination. The true-positive rate on the ROC curve is equal to C_X . The area under the ROC curve (AUC) summarizes how well a continuous score orders the two classes (T. Fawcett 2006).

4. DISCUSSION AND CONCLUSIONS

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